

Task 1 : Research Report on Robotics Applications

Title: Management And Risk
Analysis of the da Vinci
Surgical System

Abstract

This report critically assesses the da Vinci Surgical System within six management approaches: commercial and market opportunities, regulatory and professional constraints, consumer and societal attitudes, sustainability, ethical considerations and risk management. By using quantitative clinical evidence, regulatory requirements and structured analysis tools such as SWOT, Porter's Five Forces and a Risk mitigation Map, the report contends that the system's market penetration is both evidence-based and market-based, and superiority is shown in specific procedures, rather than across the board. The report demonstrates economic sustainability requires a certain volume of procedures (350+ cases per year), ruling out lower-acuity hospitals as potential adopters. The report finds the da Vinci system offers significant clinical benefit, but its governance, sustainability and equity issues are the best addressed through structural and not technological change.

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Chapter:1 Introduction

The da Vinci Surgical System (Intuitive Surgical Inc.) has been commercially available since the US Food and Drug Administration (FDA) approved it in 2000, and remains the most widely used robotic surgery system worldwide, with 8,544 systems deployed in 67 countries in 2023 (Intuitive Surgical, 2023). It allows minimally invasive surgery with robotics arms manipulated from a distant console, offering 3D high-definition visuals, scaling and tremor reduction (Lanfranco et al., 2004). This technology has been shown to deliver benefits to certain clinical outcomes- for example, blood transfusion rates drop from 22% in open prostatectomy to 1.5% with robotic surgery (Tewari et al., 2012). But the key issue this report addresses is whether the technology's popularity denotes whole-of-body benefit or a combination of clinical innovation and marketing acumen. The evidence is mixed: the system is highly beneficial for urological surgeries, more mixed in gynaecology and colorectal surgery (Vlug et al., 2011), and of little use in emergency surgery where time is critical. This is important for health care institutions making multi-million pound investments. The next sections use analytical tools to assess commercial interest, laws and regulations, user feedback, sustainability, ethics and risk – to draw a realistic, evidence-based view.



Figure 1: da Vinci Surgical System

Chapter 2: Commercial and Market Analysis

2.1 Market Dynamic: Porter's Five Forces

The surgical robotics market is a supplier-dominated market. For Intuitive, supplier power is high: Instrument licensing and service agreements are proprietary and result in future revenues that far outpace hardware profits. Buyer power is low, due to high switching costs; surgeons, operating room facilities and instrumentation interoperability lock users in once they have adopted the system. New entrants are a moderate threat; high start-up costs and regulatory hurdles exist, but Johnson & Johnson (Ottava) and Medtronic (Hugo), and CMR Surgical (Versius) are now or approaching regulatory approval (see Table 1). Threat from substitutes is low in high-complexity minimally invasive surgery, as laparoscopy is not as dexterous as robotics. Thus, rivalry is the main market variable to watch over the next five years, with price competition evident.

Criteria	Da Vinci (Intuitive)	Hugo (Medtronic)	Ottava (J&J)	Versius (CMR)	Threat Level
Market Share	~80%	Emerging	Pre-launch	<5%	–
System Cost	\$1.5-2.5M	~\$1.0-1.5M	TBC	~\$0.8-1.2M	Medium
Installed Base	8,544 units	<200	TBC	~150	Low(now)
Training Ecosystem	Extensive	Developing	Minimal	Limited	Medium
FDA/MDR Approval	Full	CE + FDA granted	In Progress	CE marked	High (5yr)

Table 1: Surgical Robotics Competitive Landscape (porter-informed, 2024)

2.2 Economic Viability and the ROI Threshold

The assessment of economic viability should not just be a statement that it's expensive. Rather, it should consider in which circumstances it is cost-effective. The cost of the da Vinci Surgical System ranges between \$1.5-2.5 million, with maintenance costs of \$150,000-180,000 a year. Moreover, disposable instruments are required for each procedure, costing around \$700-3,500. Consequently, a hospital performing approximately 200 robotics surgeries a year could end up paying \$2,000-4,000 more per procedure than a traditional laparoscopic procedure (Barbash & Glied, 2010).

While robotic surgery can increase efficiency (e.g., by shortening hospital stays by 1.2 days and decreasing complications), these cost savings are only able to offset the investment cost at high procedure volumes. For most hospitals, cost-effectiveness is achieved with more than 350 cases per year. Fewer than this, the cost is justified less by the expected return on investment and more by other factors such as brand.

The global surgical robotics market is set to increase substantially from \$7.2 billion in 2023 to \$22.9 billion in 2030 (Markets and Markets, 2023). But this expansion needs to be taken with a grain of salt. Often, hospitals adopt new technologies not just to meet clinical needs, but to position themselves as innovative health-care providers. If not managed well and not used enough, these investments become more a question of image than benefit.

Chapter 3: Professional Standards and Legal Requirements

3.1 Regulatory Framework and Compliance Burden

The da Vinci Surgical System is heavily regulated as a high-risk medical device. In the USA, it undergoes approval through the FDA Class III pre-market approval process (FDA, n.d.). In Europe, it is subjected to the Medical Device Regulation (MDR 2017/745) with a requirement for continuous monitoring, reporting, and evaluation of its safety and clinical performance after it enters the market (European Commission, 2021).

These regulations are strict due to the risks. For instance, the FDA's Manufacturer and User Facility Device Experience (MAUDE) database shows more than 10,000 reported cases of the da Vinci systems between 2000 and 2013, including fatalities and injuries (Alemzadeh et al., 2016). This shows that although robotic surgery has advantages, it is not without risks, and rigorous regulation is required.

3.2 Training Standards and Liability Architecture

A key issue with robotic surgery is the variability of training standards between institutions. It has been reported that surgeons require 150-250 supervised cases to gain proficiency (Hoekstra et al., 2017). But training programs are not always uniform, which may result in a difference in skill.

This presents safety and liability issues. Surgeons with less experience are more likely to make mistakes. Liability can fall to the surgeon, hospital and the manufacturer in these instances. It can be difficult to determine liability, particularly where it is not clear if a problem was due to human or system failure.

To minimise this, hospitals should ensure that staff are appropriately trained, accountability is clear and incidents are thoroughly investigated. This will minimise the risk of legal liability and harm to patients.

Chapter 4: Customer Satisfaction and Social Acceptance

4.1 Stakeholder Analysis

Stakeholders have varying expectations of robotic surgery. For patients, key concerns are recovery times and patient experience, and many are satisfied with the experience due to the advantages of minimally invasive surgery (Barbash & Glied, 2010).

Surgeons are often more interested in the increased dexterity, enhanced viewing capabilities and system ergonomics. But in the initial learning phase, they may have slower procedures due to getting used to the system.

Hospital administrators are more interested in throughput and utilisation, and patient appeals. At the same time, health care payers (such as health insurers) are becoming more concerned about cost-benefit, frequently using cost per quality-adjusted life year (QALY). The emphasis on value is increasingly playing a role in the uptake of robotic surgery.

4.2 Barriers and Strategic Improvement

Key barriers to social acceptance include patient anxiety about reduced direct human involvement in surgery – a perception issue rather than a clinical reality, since surgeon control is maintained throughout – and access inequality, which frames robotic surgery as a privilege of well-resourced care. Strategies to address these include:

- Competency-based credentialling aligned to the 150-250 case evidence threshold, replacing institutionally variable standards
- Patient decision aids that accurately represent the surgeon's active control, counteracting robotic 'autonomy' misconceptions
- Transparent public reporting of institutional robotic outcomes data to enable evidence-based patient choice
- Procurement models (leasing, shared access, NHS regional hubs) to expand availability beyond tertiary-funded centres

Chapter 5: Sustainability Assessment

5.1 Triple Bottom Line Framework

The Triple Bottom Line (TBL) is a helpful framework to evaluate sustainability, taking into account environmental, social and economics impacts. When it comes to the da Vinci Surgical System, the findings are mixed.

Environmentally, a primary issue is the disposable instruments. Research has shown that robotic surgery can result in approximately 30-50% more carbon emissions per case than traditional laparoscopic surgery, equating to an estimated 1,000kg of CO₂ per case (Rizan et al., 2020). Moreover, the production of the robotic system itself includes the use of advanced components such as electronics and high-precision parts, which also have an impact on the environment.

Socially, the system is advantages. It can potentially shorten patients' hospital stays and recovery periods, which can alleviate burdens on the health care system and lead to better patient outcomes. But this technology is not widely accessible and this has equity and equality implications.

From an economic perspective, the system is not sustainable unless it is used often. As noted above, the system is only cost-effective if hospitals undertake many procedures. This means that the sustainability of the system depends on its usage.

Dimension	Positive Impacts	Negative Impacts	Net Assessment
Economic	Fewer complications; 1.2-day shorter stay (prostatectomy); higher patient throughput	\$1.5-2.5M capex; \$ 150-180k/yr maintenance; breakeven ~350 + cases/yr	Viable in high-volume centres; marginal in low-volume settings
Social	Faster patient recovery; reduced surgeon fatigue; improved outcomes in prostatectomy	Access limited to well- funded hospitals; learning curve risks in early adoption	Net positive where access is equitable; ethically problematic where it is not
Environmental	Shorter stays reduce facility resource use; potential for reusable redesign	~30 – 50% higher per- procedure CO2 vs reusable laparoscopy (Rizan et al., 2020); 1,000kg CO2e/procedure estimate	Currently net negative; addressable through instrument redesign and ESG procurement policy

Table 2: Triple Bottom Line Sustainability Assessment(adapted from Elkington,1997)

At a broader influence level, some health systems (such as the National Health Service [NHS] in the UK) and organisations in the EU are beginning to include environmental factors in purchasing. This will mean that products which are wasteful or high-carbon intensive may come under further scrutiny.

In turn, manufacturers have begun to look for more environmentally friendly options, such as reusable or recyclable instruments. But this is still in its infancy, and more needs to be done to meet future requirements.

Chapter 6: Ethical Issues

6.1 Distributive Justice and the Evidence-Access Gap

A key ethical issue is not the technology itself, but distribution. Robotic surgery may offer better outcomes and faster recovery, but not everyone has access to this type of technology, which is often restricted to affluent hospitals or areas. This results in a disparity of care – some patients benefit, while others do not (Barbash & Glied, 2010).

A further problem is that much of the evidence around the clinical benefits of robotic surgery comes from types of surgery that are more prevalent in wealthier populations, like prostate surgery. This implies that the advantages of the technology are not being uniformly studied and used. This means that health-care systems need to focus not just on the effectiveness of the technology, but also its equity.

6.2 Accountability, Skill Attrition, and Data Ethics

Responsibility is also an issue. If there is an adverse event during event during robotic surgery, it may be hard to ascertain who was to blame. It may be difficult to determine whether it was the surgeon, the hospital or the manufacturer's fault, making it more difficult to hold them to account.

There is also a risk of surgeons losing their basic surgical skills. With an increasing use of the robotic system, surgeons may not get as much practice with traditional methods. This may be dangerous if the robot malfunctions and other techniques are required. So it's important for surgeons to have training in traditional surgical methods.

Lastly, robotic surgery creates a lot of data, such as video footage of surgeries and system performance data. While data can be used to improve outcomes, it can also be used to personal information and data exploitation. Robust data governance policies, patient consent and restrictions on how data is used are crucial to build trust (World Health Organization, 2021).

Chapter 7: Risk Evaluation and Management

7.1 SWOT Summary

The SWOT analysis (Table 3) provides an overview of the strengths, weaknesses, opportunities and threats of the da Vinci Surgical System, to help with the risk analysis.

Its market share and amount of clinical data are strengths. Its popularity creates a network effect which makes it difficult to dislodge. But it also has major vulnerabilities, with respect to cost and single-use instruments. This makes the system costly to operate, but also vulnerable to competition from lower cost systems.

Strengths	Weaknesses
FDA/MDR approved with over 20 years of clinical evidence	High acquisition cost (\$1.5-2.5M) limits accessibility
Large installed base (8,544 units) creating strong network effect	Long learning curve (150-250 cases to achieve proficiency) (Hoekstra et al., 2017)
Proprietary instrument ecosystem ensures continuous revenue	Single-use instruments increase per-procedure cost (\$700-3,500)
Tremor elimination reduces complications reduces complication risk by up to 21% (Lanfranco et al., 2004)	Clinical superiority not proven in all surgical specialities
Lower blood transfusion rates compared to traditional surgery	Proprietary system limits hospital bargaining power

Opportunities	Threats
Rapid market growth (CAGR 17.9% from \$7.2B in 2023 to \$22.9B by 2030)	Increasing competition from Medtronic and Johnson & Johnson
Integration of AI for improved surgical guidance	Reported adverse events in FDA MAUDE database (Alemzadeh et al., 2016)
Expansion into emerging markets (Asia, Africa)	Cybersecurity risks due to network connectivity
Growth of telesurgery and remote procedures	Stricter regulatory requirements (MDR compliance)
Development of reusable instruments to improve sustainability	Public concern over cost-effectiveness and ROI

Table 3: SWOT Analysis Overview

7.2 Risk Assessment and Mitigation Framework

The risk assessment (Table 4) identifies and rates the risks based on their likelihood and impact. Some risks are critical, such as system failure during surgery and not meeting regulatory requirements. These risks must be managed at all times because of their impact on the patient, health care provider reputation and the health care industry. There are other risks such as human error and cybersecurity risks. These require strategies such as training, monitoring and security.

Risk	Likelihood	Impact	Rating	Primary Mitigation
Intraoperative system failure	Low	Catastrophic	Critical	Redundant systems; open-surgery fallback protocol
Human error (learning curve)	Medium	Severe	High	Competency-based training; 150+ supervised cases
Financial underperformance (ROI)	Medium	Significant	Medium	Volume threshold planning; leasing models
Cybersecurity breach	Low	Severe	High	Network security measures; penetration testing
Regulatory non-compliance	Low	Catastrophic	Critical	Dedicated compliance team; MDR/FDA monitoring
Competitive obsolescence	Medium	Moderate	Medium	Investment in innovation; technology upgrades

Table 4: Risk Assessment and Mitigation

The FDA/MAUDE database shows many issues arise during surgery, due to system failure (Alemzadeh et al., 2016). This suggests operational risks ought to be taken into account.

The next issue is cybersecurity. There is a growing risk that systems may be hacked. The risk may be low but the consequences could be high. It's therefore better to be proactive and ensure security rather than reactive and deal with issues as they occur.

7.3 Overall Risk Assessment

Overall, the risk exposure to the da Vinci system can be managed but it will require support planning and support. This includes training, maintenance, compliance and security.

If these additional requirements are not considered, and the hospital only considers the initial purchase of the system, they may underestimate the cost. This could lead to poor cost management and lost value.

Chapter 8: Conclusion

The da Vinci Surgical System is an innovative and valuable addition to the field of surgery, offering benefits in terms of precision, patient outcomes and recovery. But the advantages are not universal and depend on the procedure.

The system is best suited for high-volume centres for commercial reasons. From a clinical point of view, although there is evidence for some surgeries this is not a given.

Sustainability is a problem, due to disposable instruments and so on. There are other ethical issues such as access, accountability and privacy.

The dangers of the system can be minimised, but it's not as simple as it appears. So, it's not just the system, but the planning, governance and access to the system that are important.

So, the da Vinci Surgical System is wonderful, but it will only be successful if due account is taken of economics, ethical and operational considerations, as well as technical.

References

- Alemzadeh, H., Raman, J., Leveson, N., Kalbarczyk, Z., & Iyer, R. K. (2016). Adverse events in robotic surgery: A retrospective study of 14 years of FDA data. *PLOS ONE*, 11(4), e0151470.
- Barbash, G. I., & Glied, S. A. (2010). New technology and health care costs – the case of robot assisted surgery. *New England Journal of Medicine*, 363(8), 701-704.
- Elkington, J. (1997). *Cannibals with forks: The triple bottom line of 21st century business*. Capstone.
- European Commission. (2021). *Medical Device Regulation (EU) 2017/745*.
- Food and Drug Administration. (n.d.). *Surgical robots*.
- Hoekstra, L. T., Buyne, O. R., Ek, A., Nieveen van Dijkum, E. J. M., & Boermeester, M. A. (2017). Learning curve in robotic-assisted laparoscopic surgery. *Journal of Laparoendoscopic and Advanced Surgical Techniques*, 27(8), 775-781.
- Intuitive Surgical. (2023). *2023 annual report*. Intuitive Surgical Inc.
- Lanfranco, A. R., Castellanos, A. E., Desai, J. P., & Meyers, W. C. (2004). Robotic surgery: A current perspective. *Annals of Surgery*, 239(1), 14-21.
- MarketsandMarkets. (2023). *Surgical robots market – global forecast to 2030*. MarketsandMarkets Research.
- Rizan, C., Bhatt, R., & Bhatt, D. (2020). Environment impact of surgical robots: A systematic review. *Journal of Cleaner Production*, 272, 122864.
- Tewari, A., Sooriakumaran, P., Bloch, D. A., Seshadri-Kreaden, U., Herbert, A. E., & Wiklund, P. (2012). Positive surgical margin and perioperative complications rates of primary surgical treatments for prostate cancer. *European Urology*, 62(1), 1-15.
- Vlug, M. S., Wind, J., Hollmann, M. W., Ubbink, D. T., Cense, H. A., & Bemelman, W. A. (2011). Laparoscopy in combination with fast track multimodal management is the best perioperative strategy in patients undergoing colonic surgery. *Annals of Surgery*, 254(6), 868-875.
- World Health Organization. (2021). *Ethics and governance of artificial intelligence for health*. WHO Press.